Physical Chemistry (I)

Lecture 14

Liquid-Gas Transition

In the Last Class

- Chemical potentials
  - Chemical potentials
  - Meanings
- Thermodynamic potentials
  - Modified differential forms
  - Gibbs-Duhem equation
- Phase diagram and phase rule

Phase Transition

Spontaneous conversion

of one phase into another phase

- occurs at transition temperature T_{trs} (for given p)
- At T_{trs}, the two phases are at equilibrium

Last class

Three Types of Equilibria

$$\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V + \mu\mathrm{d}n$$

- Mechanistic equilibrium: $p_1 = p_2$
  - Conjugate extensive variable: dV
- Thermal equilibrium: $T_1 = T_2$
  - Conjugate extensive variable: $dS = \delta q/T$
- Diffusive equilibrium: $\mu_1 = \mu_2$
  - Conjugate extensive variable: dn

μ as Molar Gibbs Energy

$$\mu = \left(\frac{\partial G}{\partial n}\right)_{T,p} = \frac{G}{n}$$

$$G = \mu n$$

$$A = \mu n + TS$$

$$H = \mu n - pV$$

$$U = \mu n + TS - pV$$

Gibbs-Duhem Equation

$$U = \mu n + TS - pV$$

$$S\mathrm{d}T - V\mathrm{d}p + n\mathrm{d}\mu = 0$$

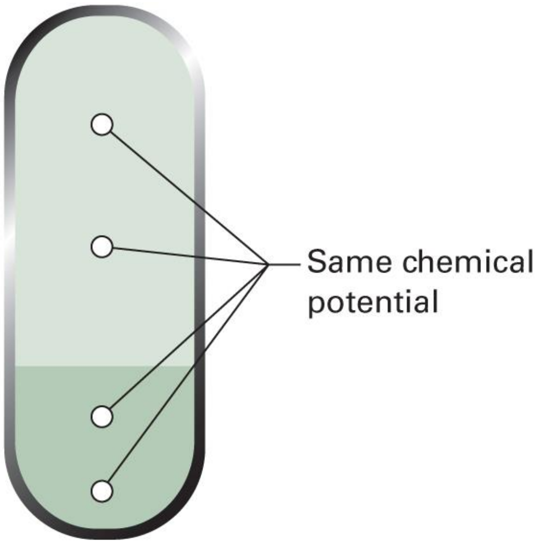
(Gibbs-Duhem equation)

$$\mathrm{d}\mu = -S_{\mathrm{m}}\mathrm{d}T + V_{\mathrm{m}}\mathrm{d}p$$

Last class

Criteria for Phase Stability

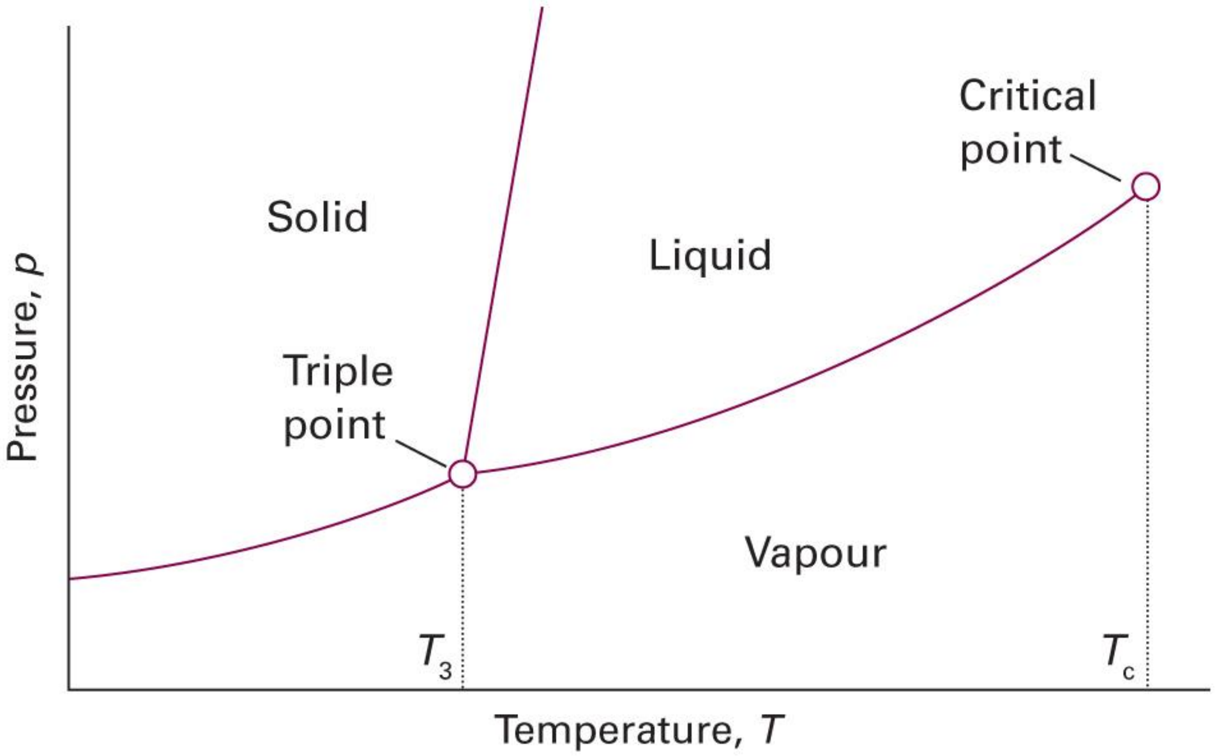
At equilibrium, the chemical potential of a substance is the same in and throughout every phase in the system



(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 11th ed., Figures 4A.3)

Last class

Phase Diagram



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figures 4A.4)

Phase Rule

"Can we have a quadruple point?"

- Phase rule: a general relation between
  - Number of degrees of freedom F
  - Number of components C
  - Number of phases at equilibrium P

$$F = C - P + 2$$

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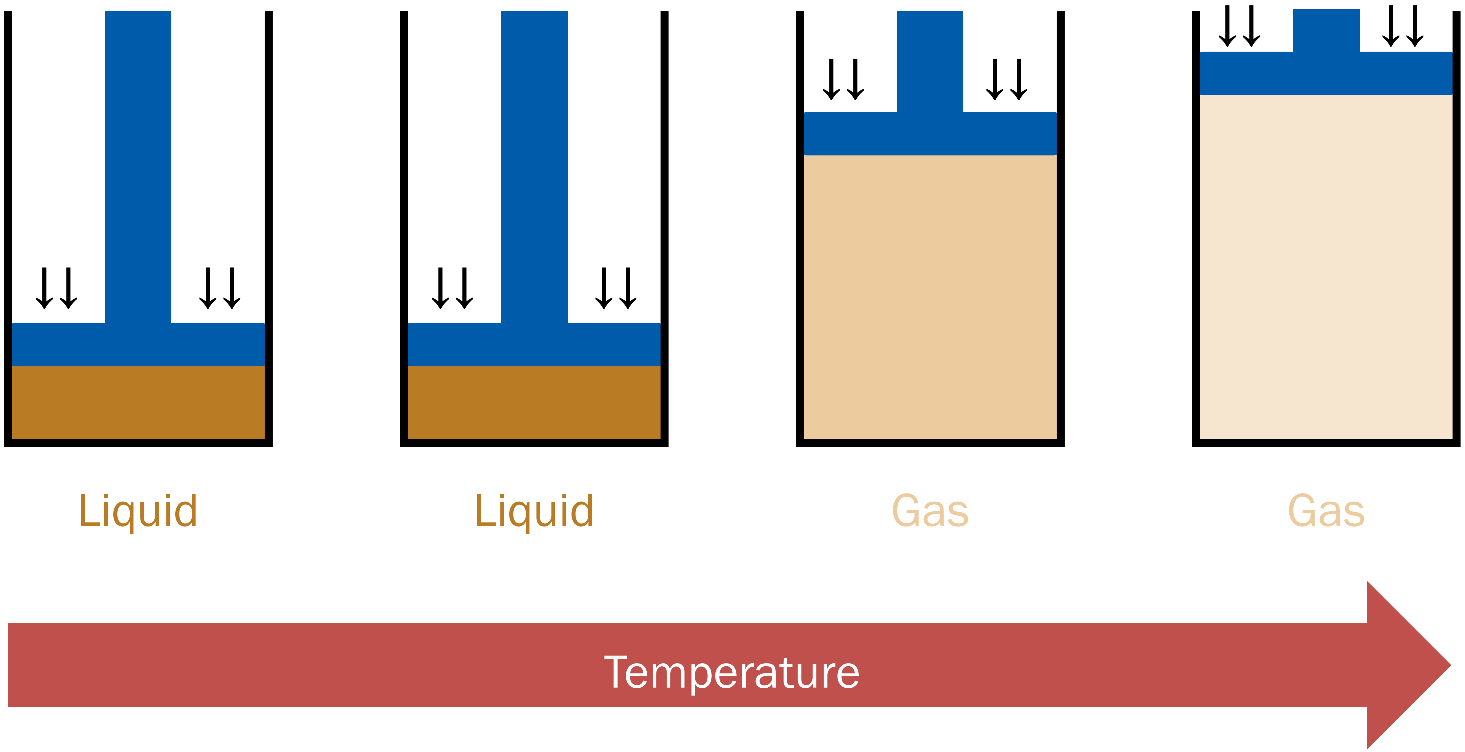
Liquid-Gas Transition 1

(Isobaric change)



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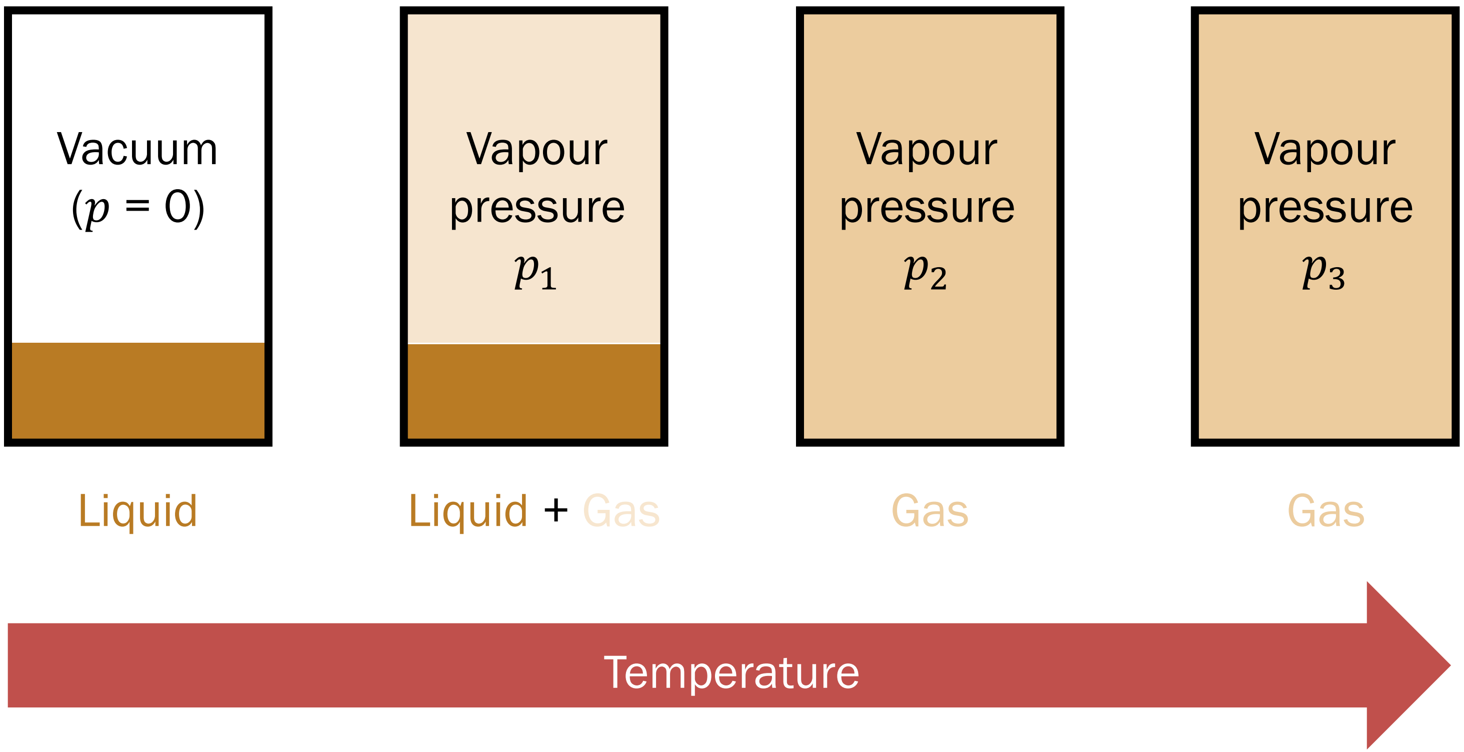
Liquid-Gas Transition 2

(Isochoric change)

$(p_1 < p_2 < p_3)$



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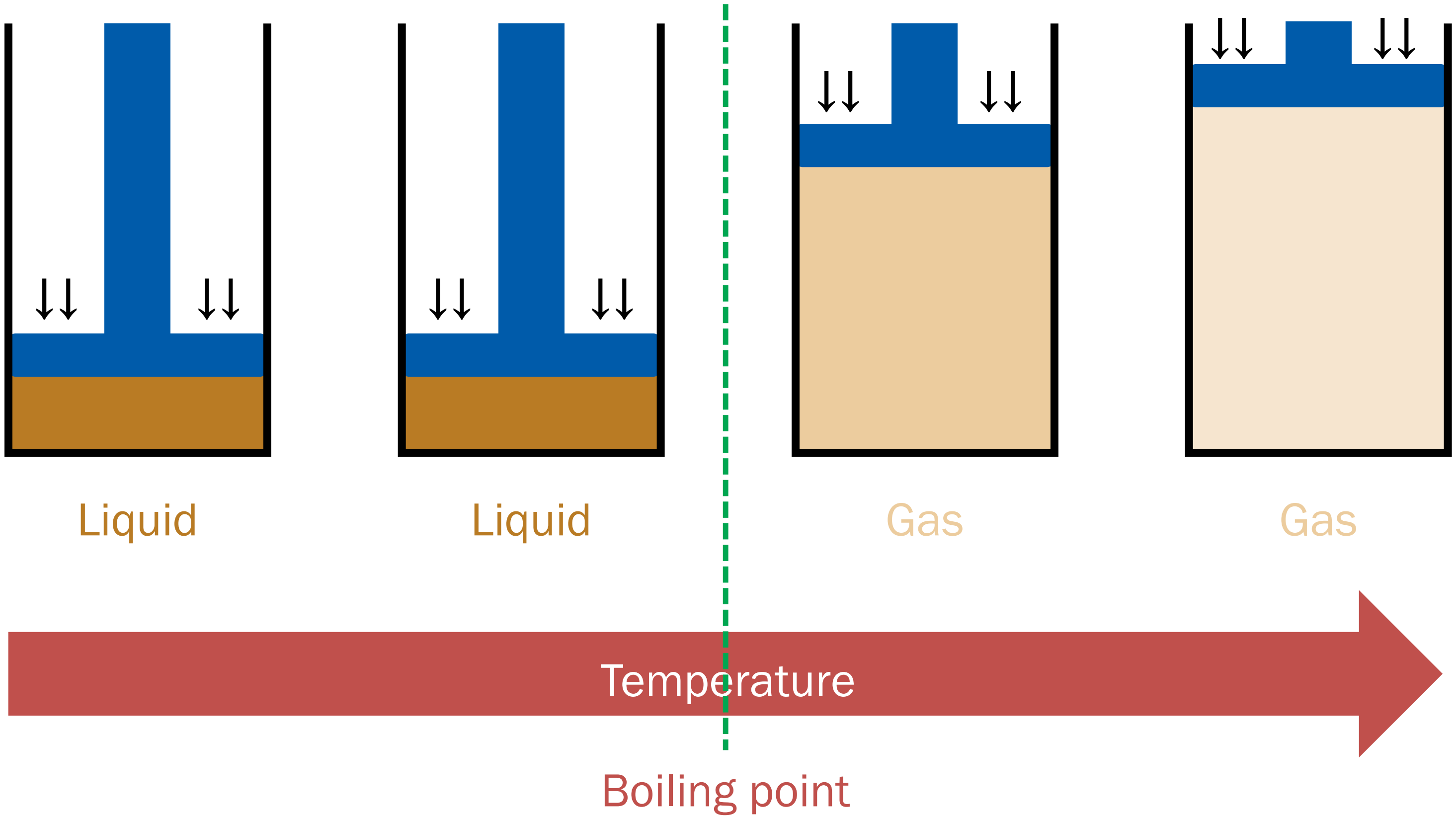
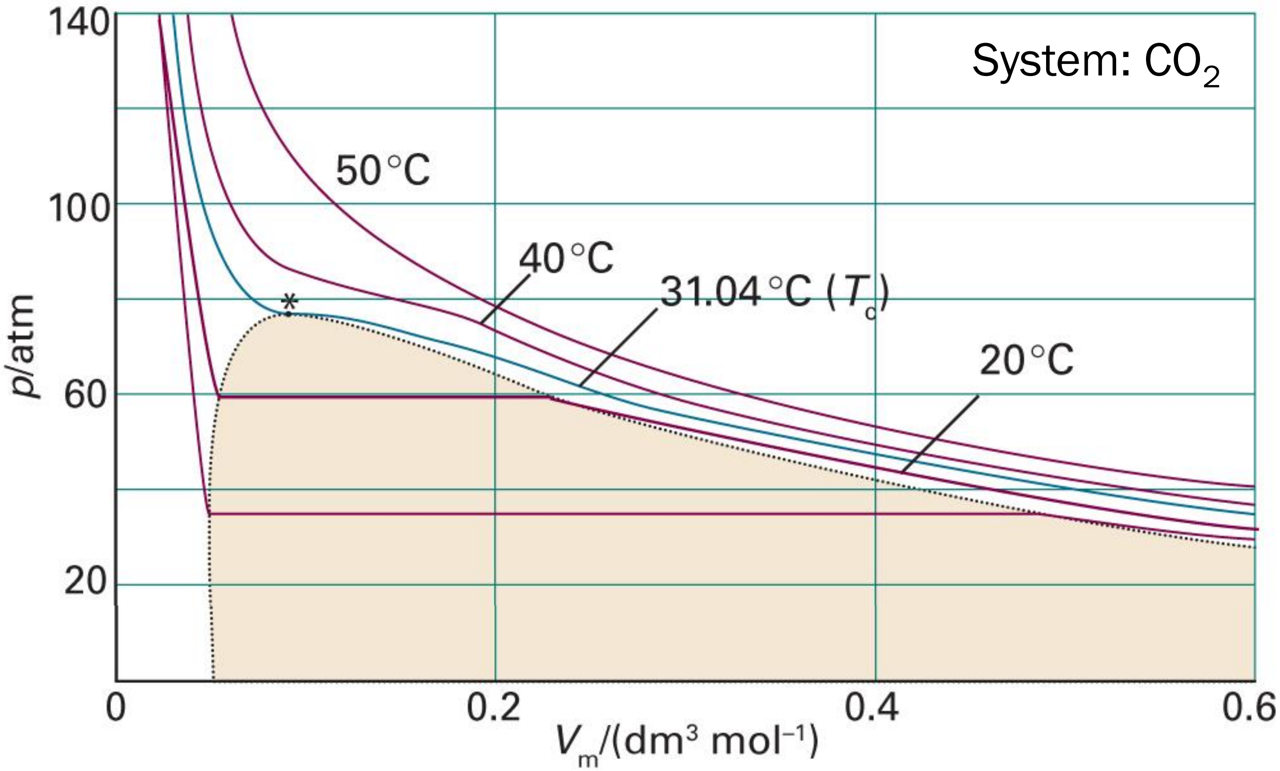
Vapor Pressure and Boiling

(Isobaric change)



Liquid-Gas Transition 3

(Isothermal change)



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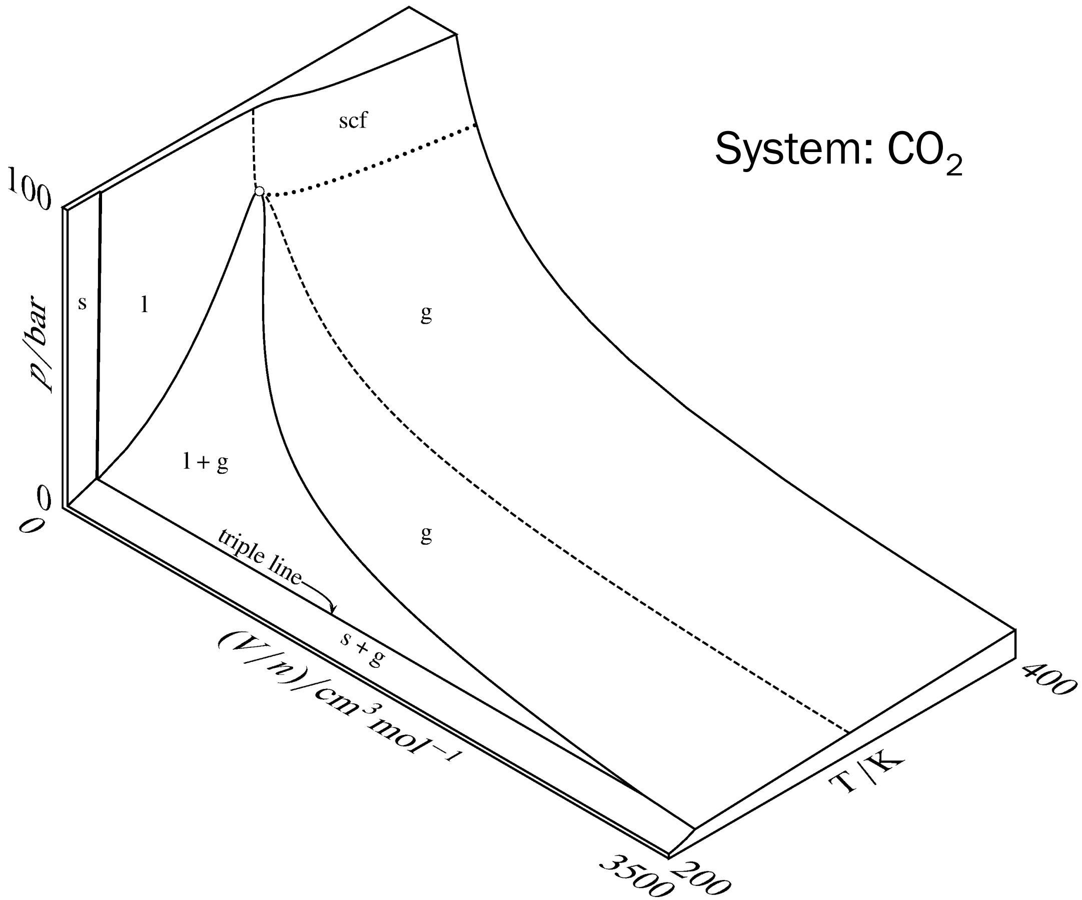
3-Dimensional Surface



System: CO_2

(Image: https://chem.libretexts.org/@go/page/20608)

Good Old Friend

"Can we describe the liquid-gas transition using a model system?"

The van der Waals fluid!

$$p = \frac{nRT}{V - nb} - a\frac{n^2}{V^2}$$

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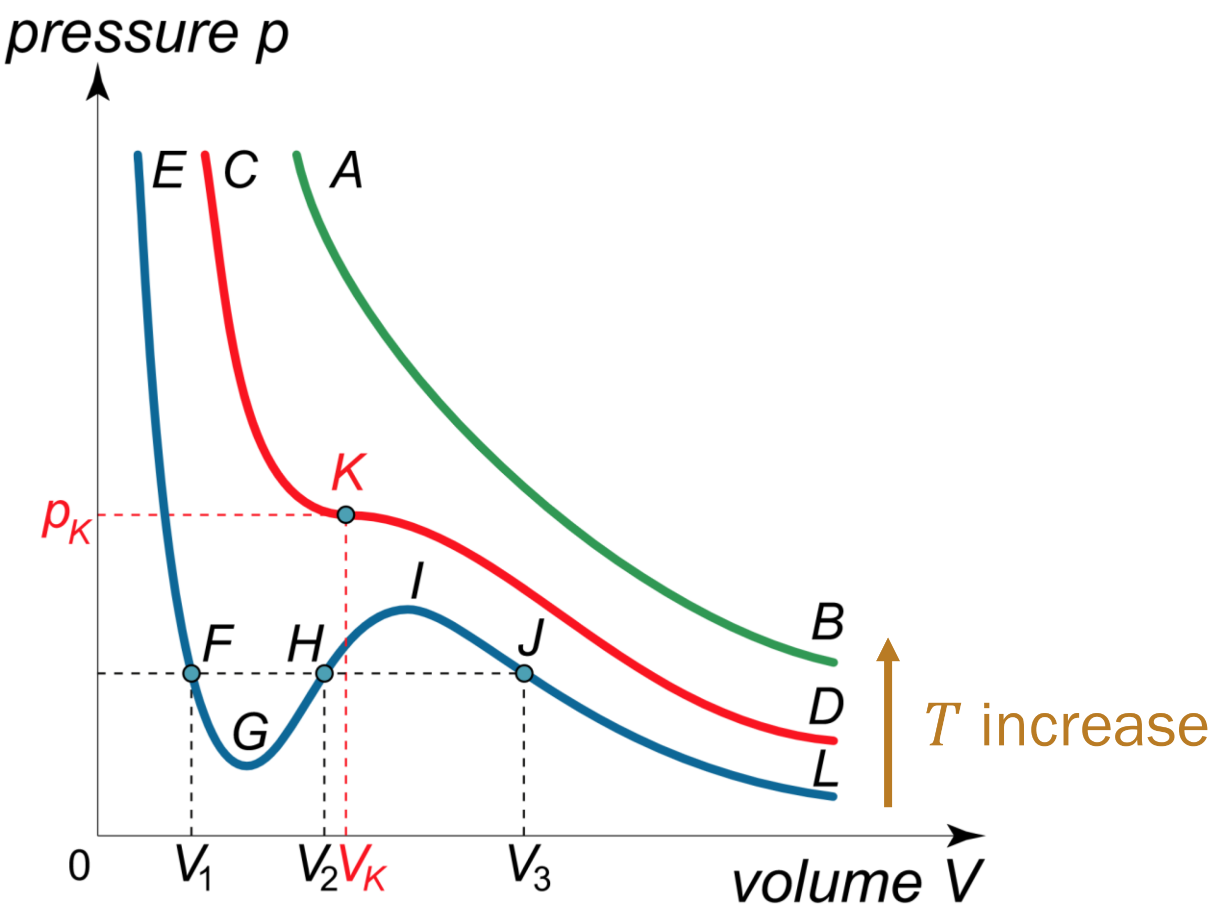
van der Waals Isotherms



(Image: https://www.math24.net/wp-content/uploads/2018/11/isoterms-of-van-der-waals-equation.svg)

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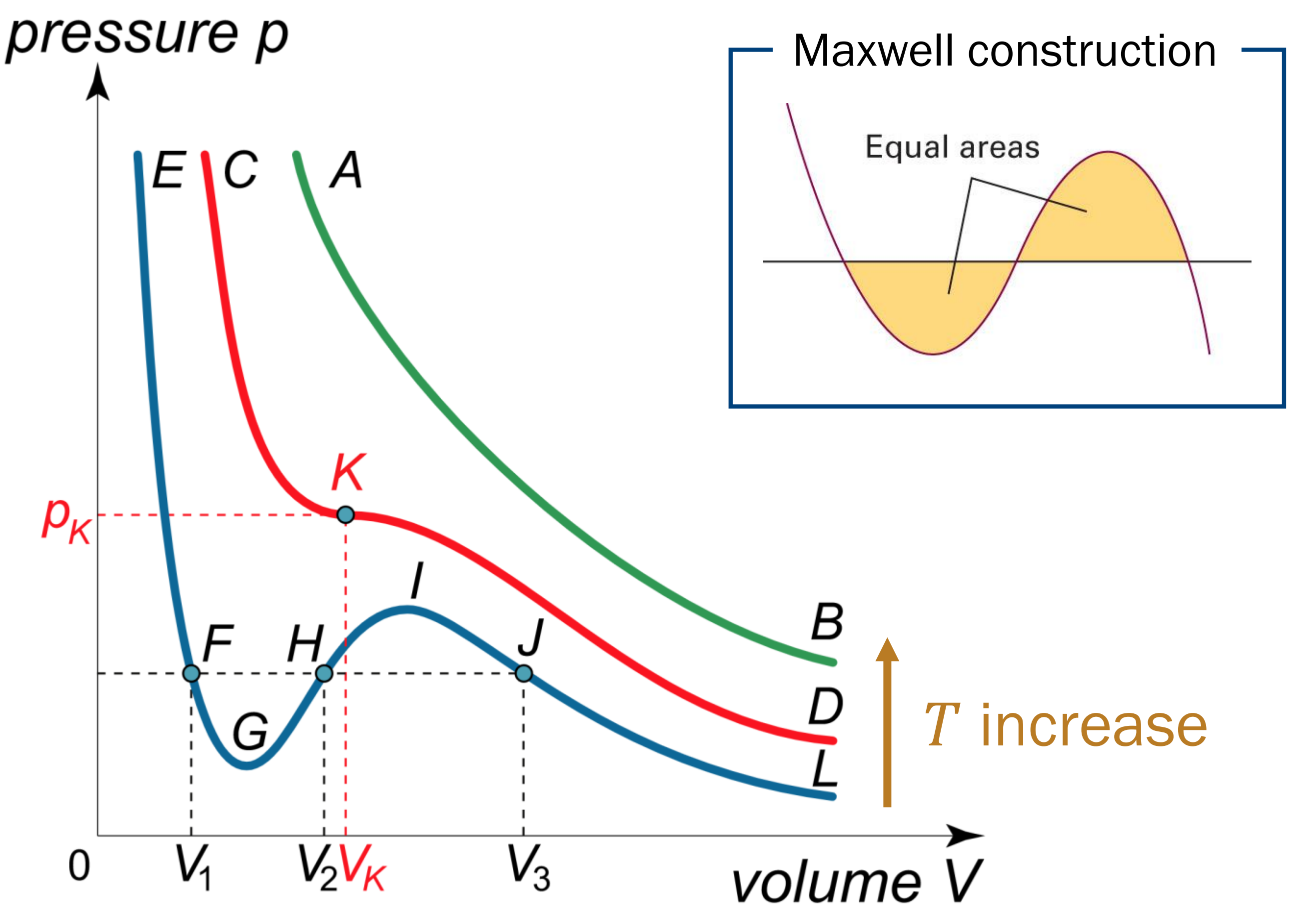
Curve E-L: Liquid-Gas Transition



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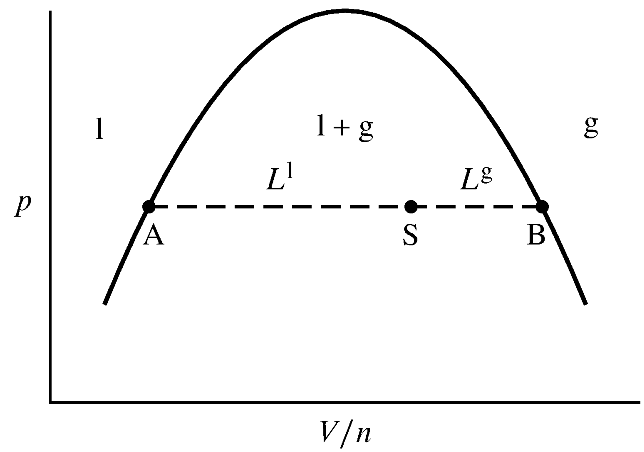
What Happens on Tie Line?



- Total amount at point S: $n = n_{\mathrm{l}} + n_{\mathrm{g}}$
- Total volume at point S: $V = V_{\mathrm{m,l}} n_{\mathrm{l}} + V_{\mathrm{m,g}} n_{\mathrm{g}}$

$$\frac{V}{n} = \frac{V_{\mathrm{m,l}} n_{\mathrm{l}} + V_{\mathrm{m,g}} n_{\mathrm{g}}}{n_{\mathrm{l}} + n_{\mathrm{g}}}$$

(Image: https://chem.libretexts.org/@go/page/20608)

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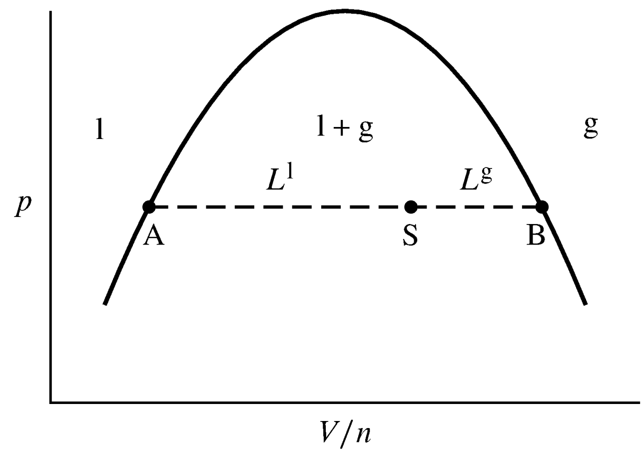
What Happens on Tie Line?



$$n_l\left(V_{m,l} - \frac{V}{n}\right) = n_g\left(\frac{V}{n} - V_{m,g}\right)$$

$$n_l L^l = n_g L^g$$

$$n_l : n_g = L^g : L^l$$ (lever rule)

(Image: https://chem.libretexts.org/@go/page/20608)

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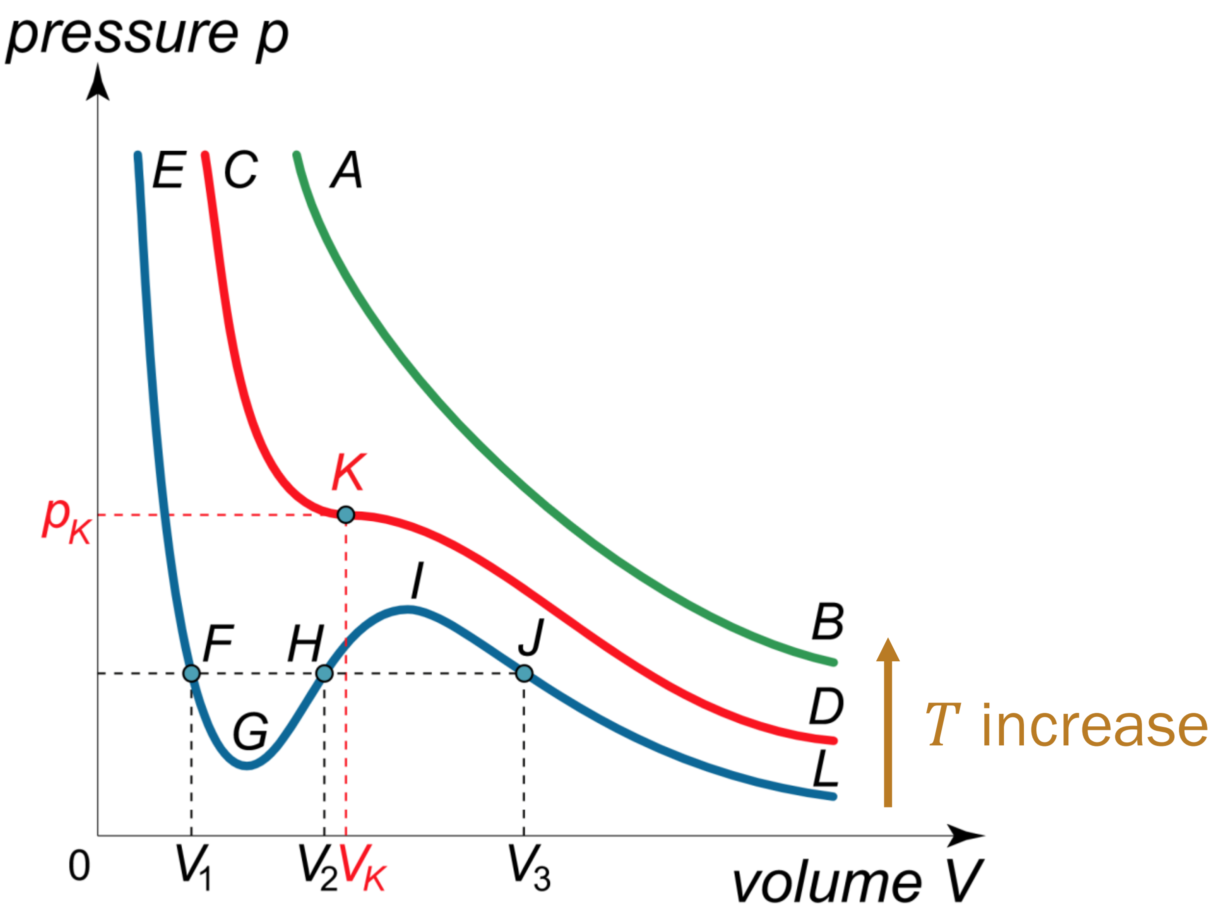
F-G and I-J: What Happens?



(Image: https://www.math24.net/wp-content/uploads/2018/11/isoterms-of-van-der-waals-equation.svg)

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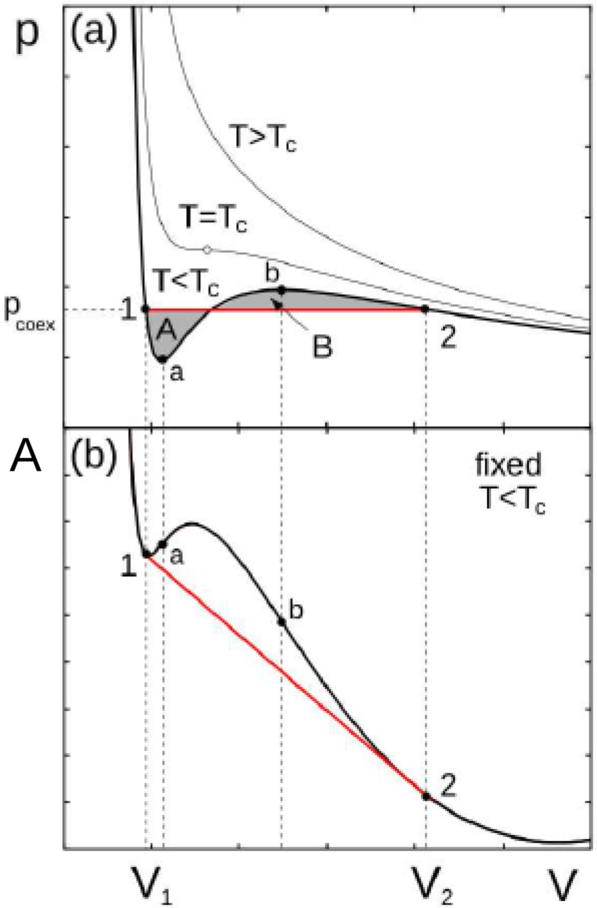
Metastable Phases

$$p = -\left(\frac{\partial A}{\partial V}\right)_{T,n}$$

At constant T and n,

$$A = -\int p\,\mathrm{d}V$$



(Image: https://www.physicsforums.com/threads/liquid-gas-phase-transition-metastable-mixed-states.935905)

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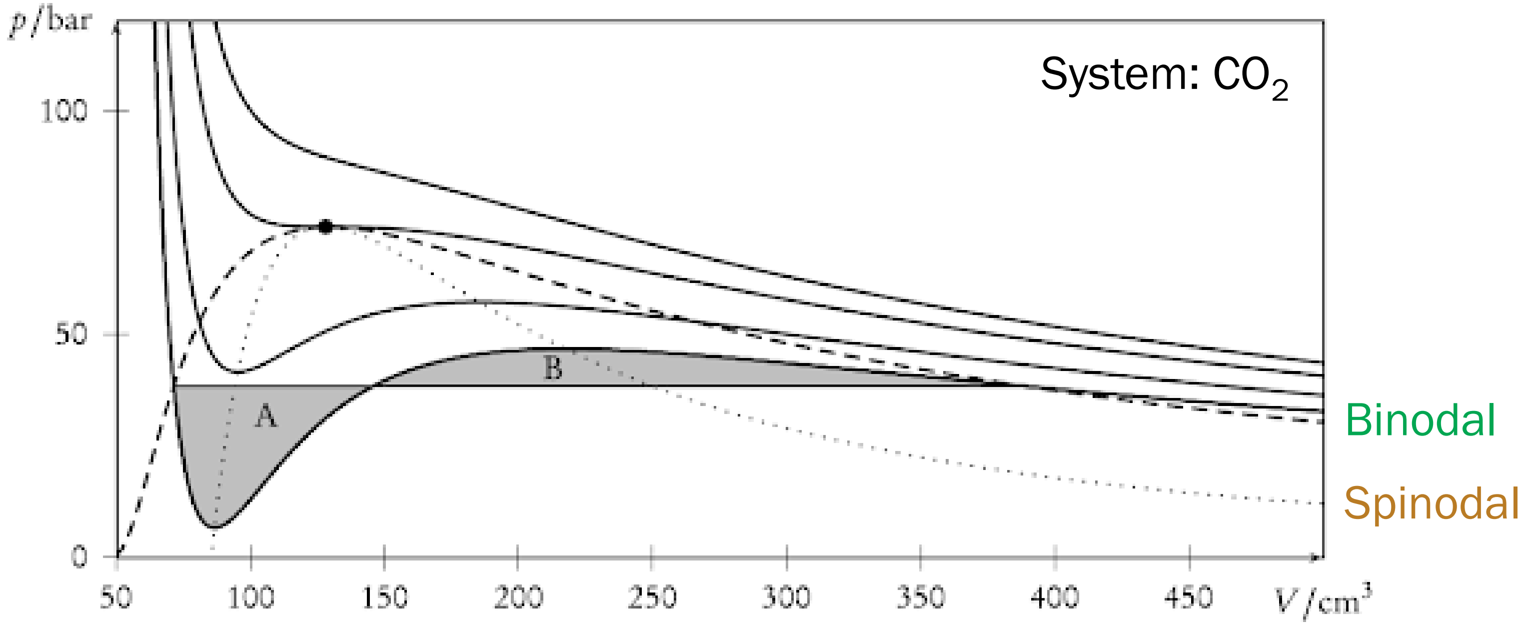
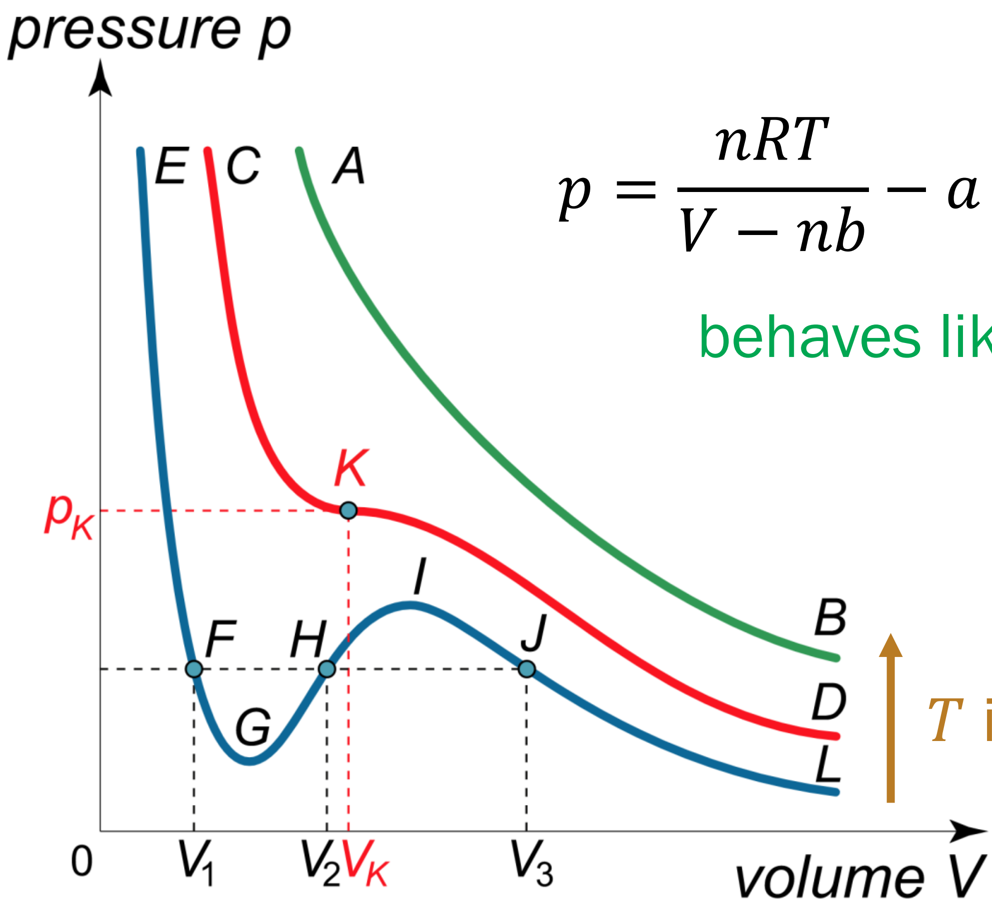
Binodal and Spinodal Lines



(Image: http://www.mhkoepf.de/Notes/Spinodal/spinodal.html)

Curve A-B: Supercritical Fluid



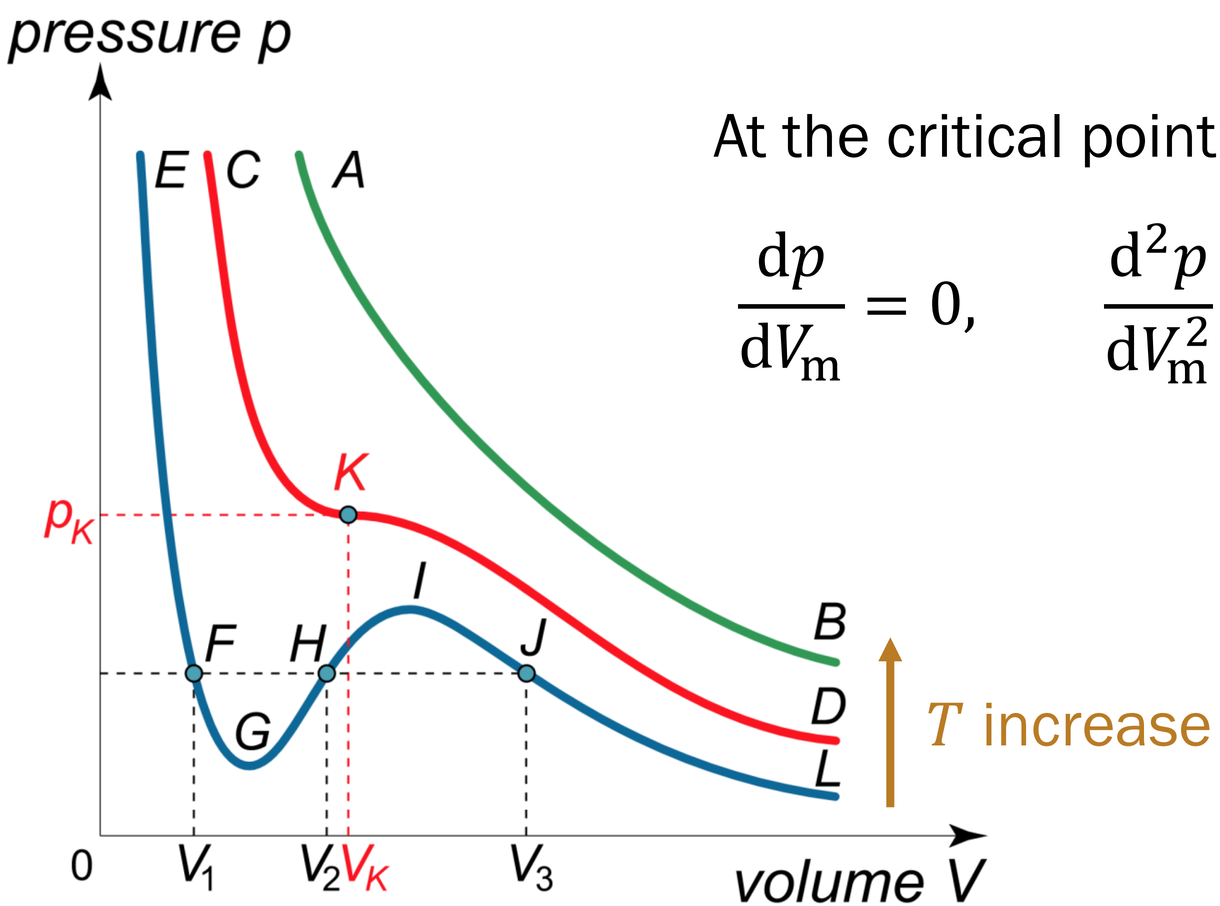
$$p = \frac{nRT}{V-nb} - a\frac{n^2}{V^2} \approx \frac{nRT}{V-nb} \approx \frac{nRT}{V}$$

behaves like a perfect gas

T increase

(Image: https://www.math24.net/wp-content/uploads/2018/11/isoterms-of-van-der-waals-equation.svg)

Curve C-D: Critical Point



At the critical point (K),

$$\frac{\mathrm{d}p}{\mathrm{d}V_\mathrm{m}} = 0, \qquad \frac{\mathrm{d}^2 p}{\mathrm{d}V_\mathrm{m}^2} = 0$$

(Image: https://www.math24.net/wp-content/uploads/2018/11/isoterms-of-van-der-waals-equation.svg)

Critical Constants

$$p = \frac{nRT}{V - nb} - a\frac{n^2}{V^2} = \frac{RT}{V_\mathrm{m} - b} - \frac{a}{V_\mathrm{m}^2}$$

$$\frac{\mathrm{d}p}{\mathrm{d}V_\mathrm{m}} = -\frac{RT}{(V_\mathrm{m} - b)^2} + \frac{2a}{V_\mathrm{m}^3} = 0$$

$$\frac{\mathrm{d}^2p}{\mathrm{d}V_\mathrm{m}^2} = \frac{2RT}{(V_\mathrm{m} - b)^3} - \frac{6a}{V_\mathrm{m}^4} = 0$$

Solving these equations, one can obtain

$$V_\mathrm{c} = 3b, \qquad p_\mathrm{c} = \frac{a}{27b^2}, \qquad T_\mathrm{c} = \frac{8a}{27bR}$$

Critical Compression Factor

$$V_c = 3b, \qquad p_c = \frac{a}{27b^2}, \qquad T_c = \frac{8a}{27bR}$$

The critical compression factor is defined as

$$Z_c = \frac{p_c V_c}{RT_c} = \frac{3}{8} = 0.375$$

Experimental values of Z_c:

Ar – 0.292, CO_2 – 0.274, He – 0.305, O_2 – 0.308

Reduced Variables

$$p_\mathrm{r} = p/p_\mathrm{c}$$

$$V_\mathrm{r} = V_\mathrm{m}/V_\mathrm{c}$$

$$T_\mathrm{r} = T/T_\mathrm{c}$$

Variables without the substance-specific features

Rewriting vdW Equation

$$p = \frac{RT}{V_\mathrm{m} - b} - \frac{a}{V_\mathrm{m}^2}$$

$$p_\mathrm{r} p_\mathrm{c} = \frac{RT_\mathrm{r}T_\mathrm{c}}{V_\mathrm{r}V_\mathrm{c} - b} - \frac{a}{V_\mathrm{r}{}^2 V_\mathrm{c}^2}$$

$$\frac{ap_\mathrm{r}}{27b^2} = \frac{8aT_\mathrm{r}}{27b(3bV_\mathrm{r} - b)} - \frac{a}{9b^2V_\mathrm{r}^2}$$

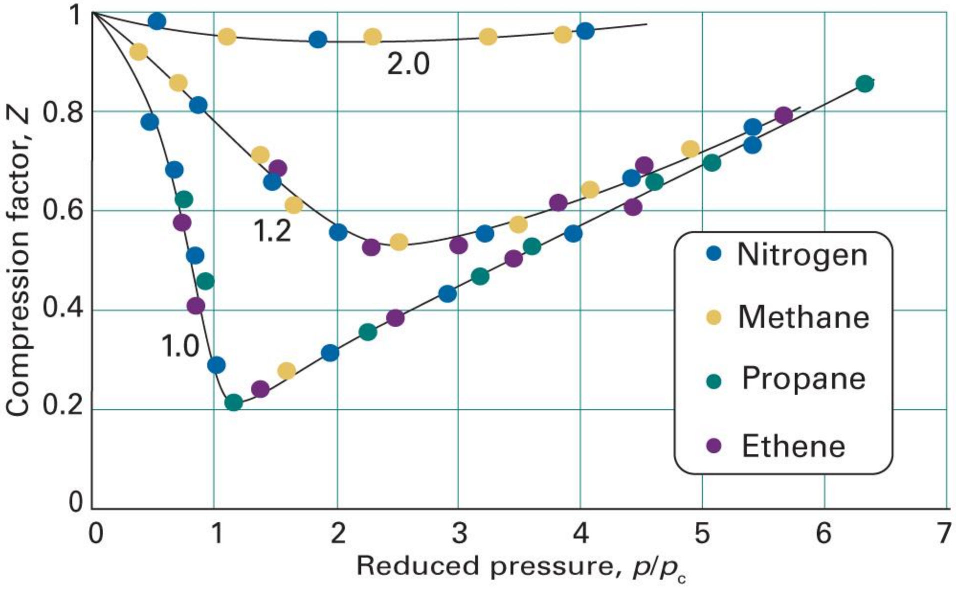
$$p_\mathrm{r} = \frac{8T_\mathrm{r}}{3V_\mathrm{r} - 1} - \frac{3}{V_\mathrm{r}^2}$$

Topic 1C

Principle of Corresponding States

Gases at the same V_r and T_r exert the same p_r

$$Z = \frac{pV_m}{RT}$$



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figures 1C.8)

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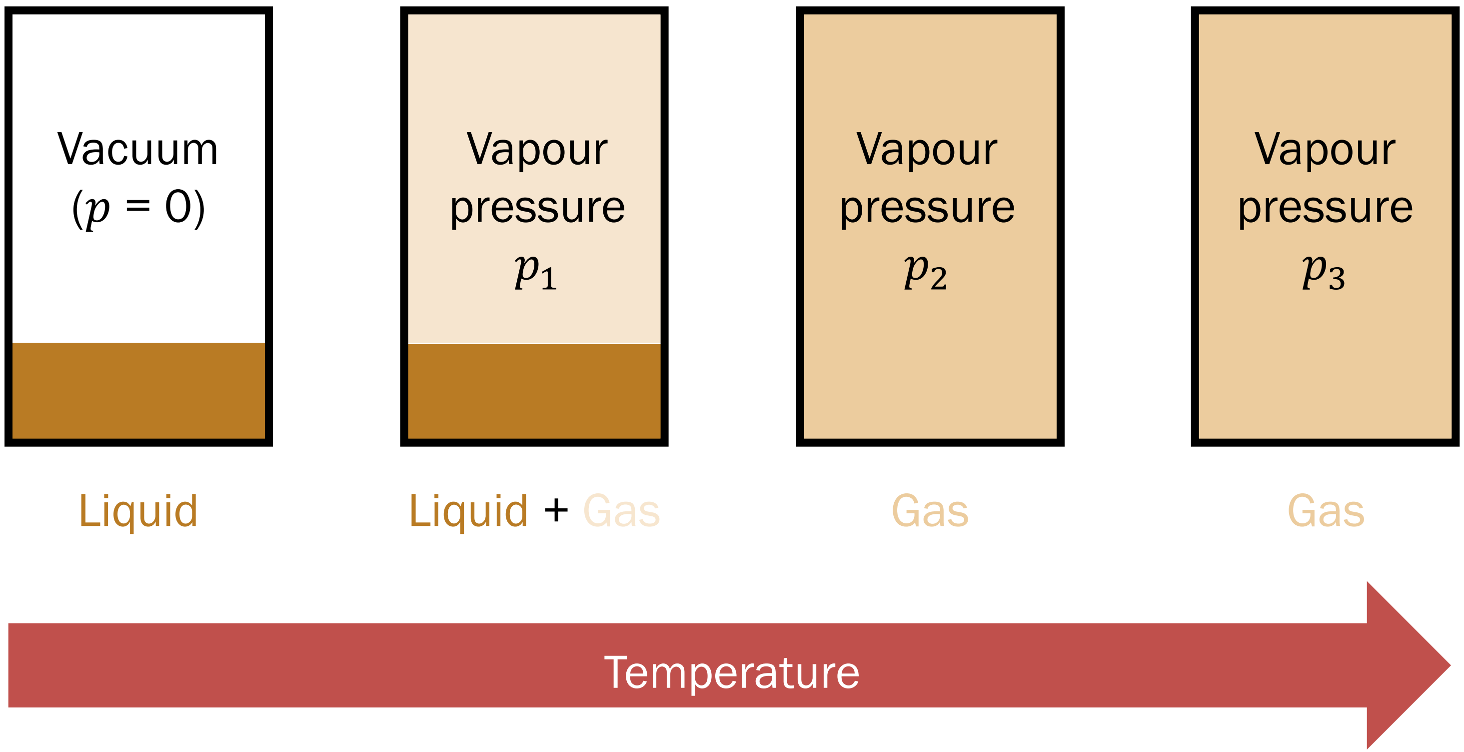
Coexistence Region

(Isochoric change)

$(p_1 < p_2 < p_3)$



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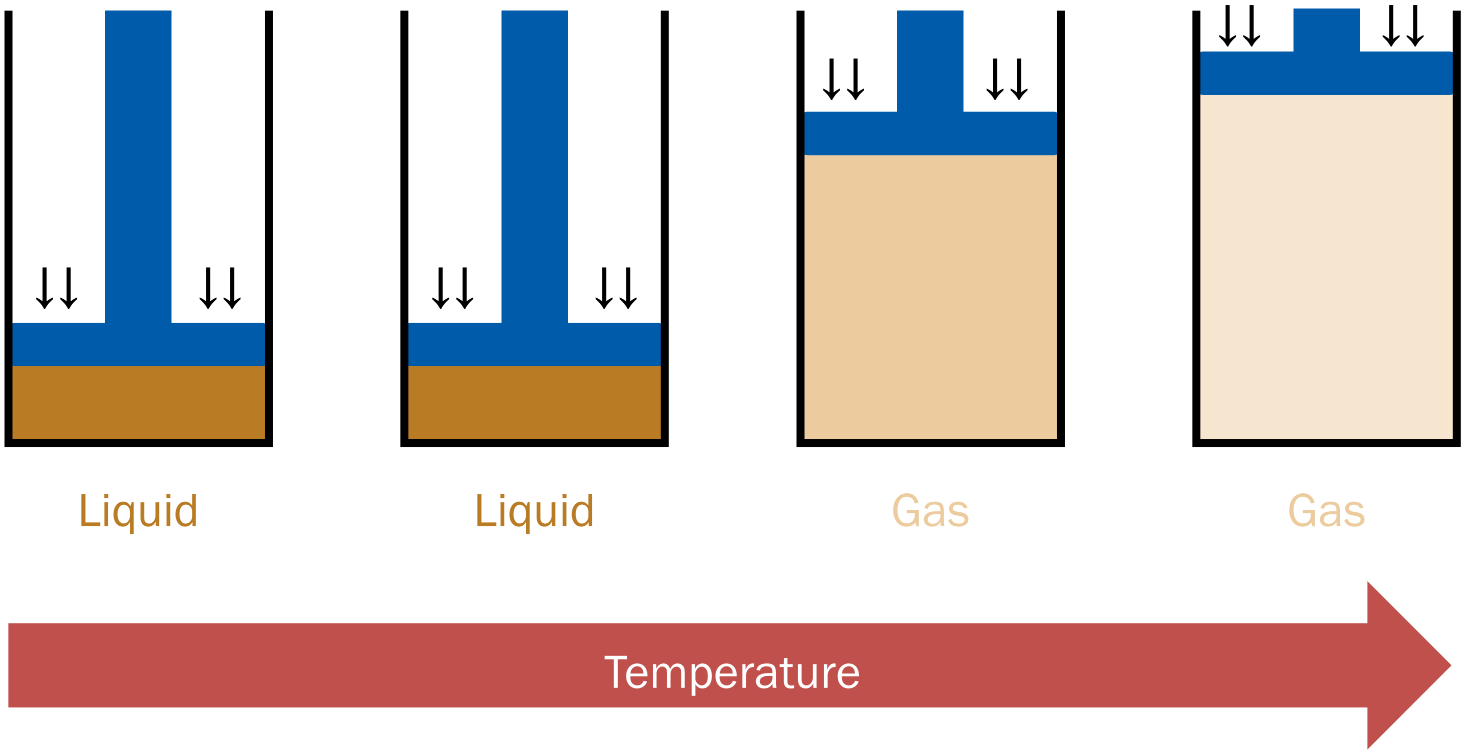
No Coexistence Region

(Isobaric change)



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Transitions of vdW Fluid

Fixed	Independent	Dependent	Coexistence?
V	T	p	Yes
p	T	V	No
T	V	p	Yes
T	p	V	?

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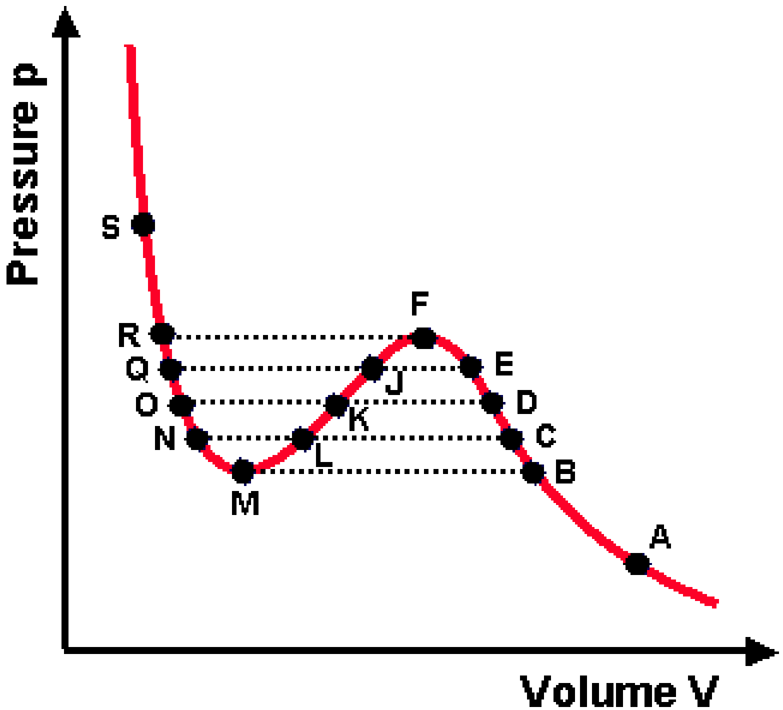
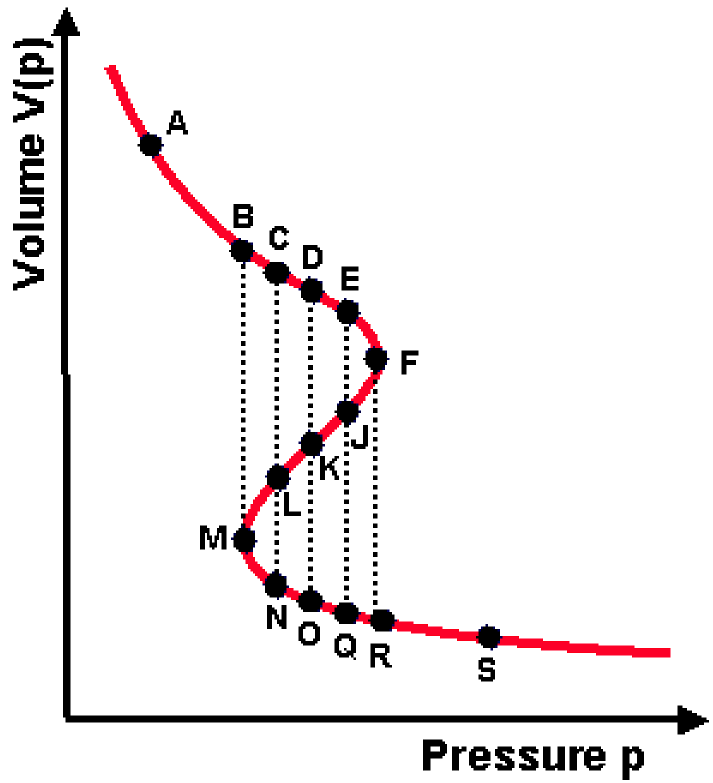
From $p(V)$ to $V(p)$





(Image: http://casey.brown.edu/chemistry/misspelled-researh/crp/Edu/Documents/00_Chem201/7_phase_equil_1/Chem201_7_phase_equil_1.htm)

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Gibbs Energy of vdW Fluid

$$V = \left(\frac{\partial G}{\partial p}\right)_{T,n}$$

At constant T and n,

$$G = \int V \, \mathrm{d}p$$

or,

$$\mu = \int V_{\mathrm{m}} \, \mathrm{d}p$$

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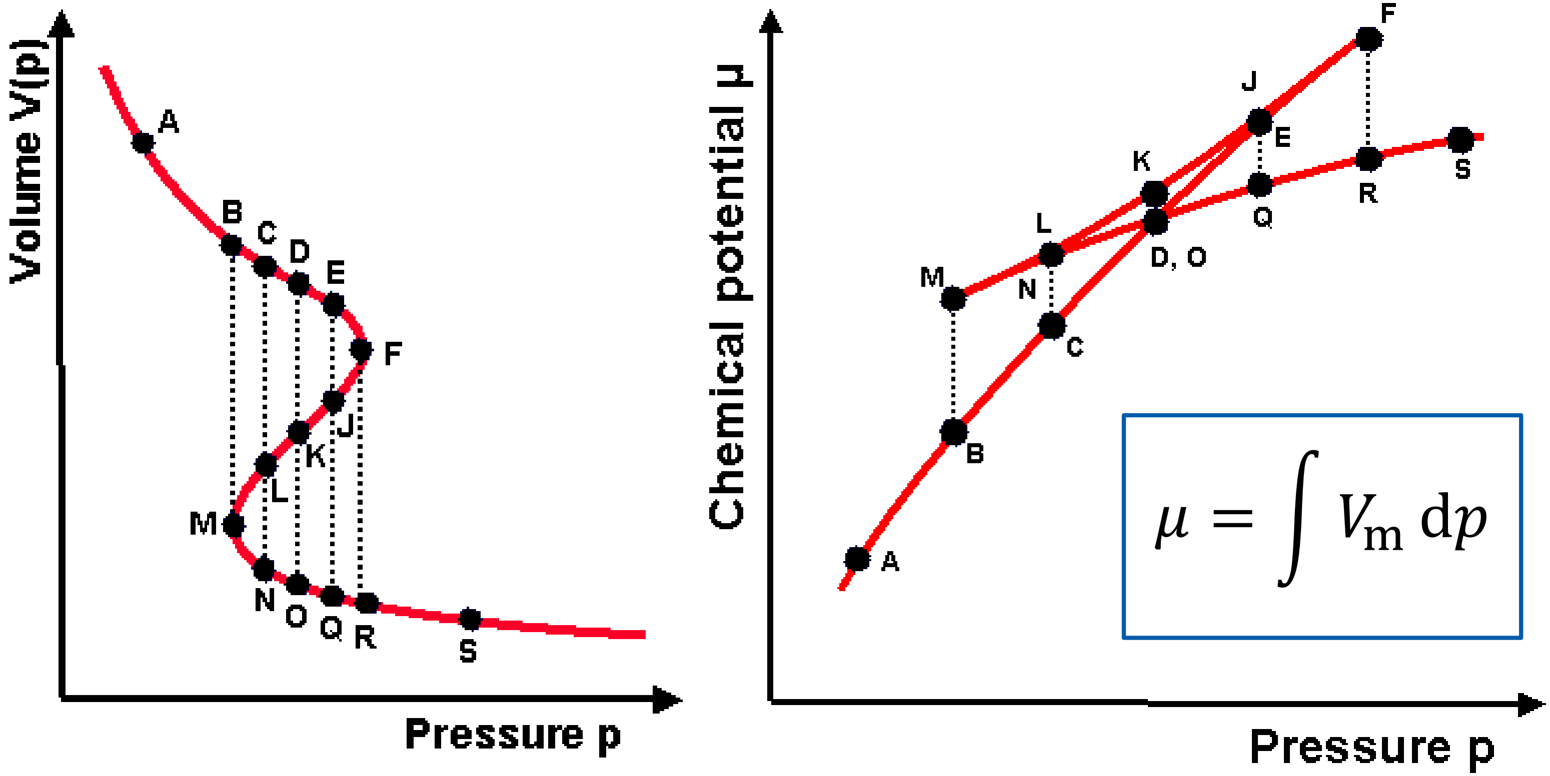
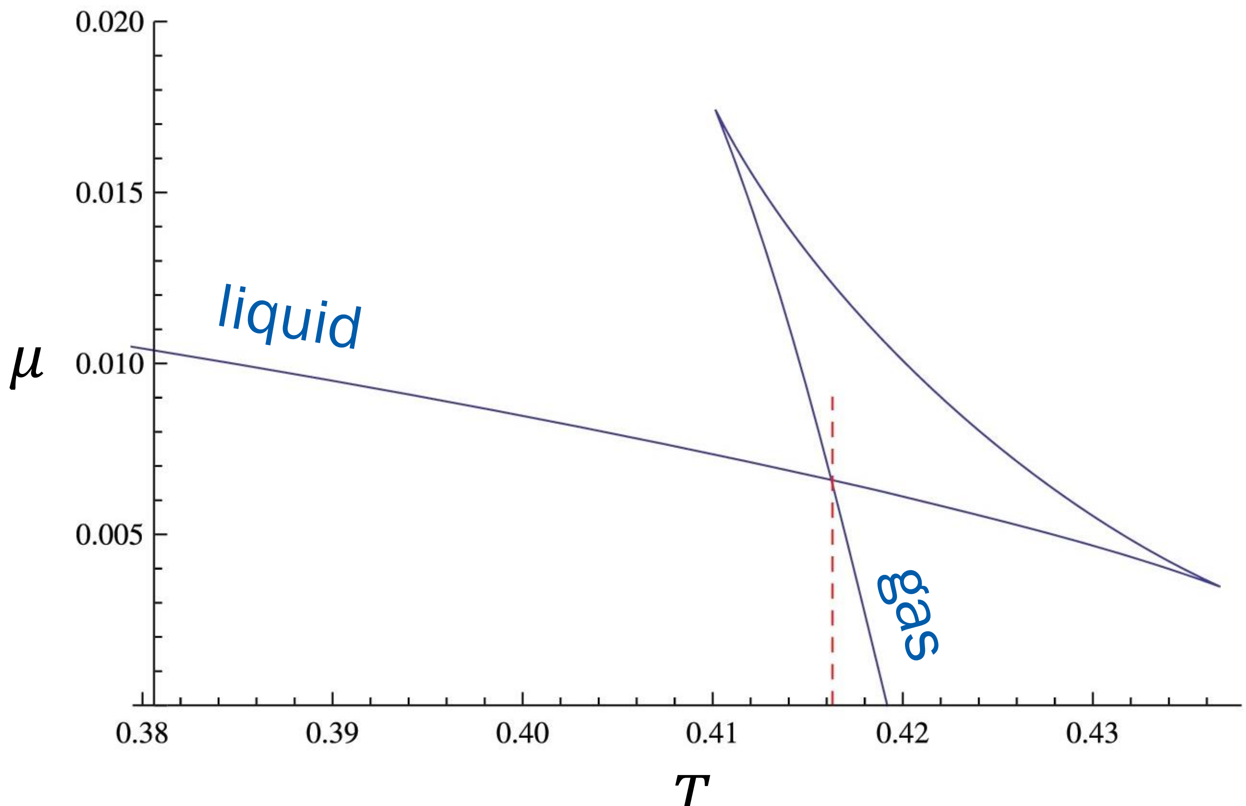
Chemical Potential $\mu(p)$



$$\mu = \int V_\mathrm{m}\,\mathrm{d}p$$

(Image: http://casey.brown.edu/chemistry/misspelled-researh/crp/Edu/Documents/00_Chem201/7_phase_equil_1/Chem201_7_phase_equil_1.htm)

$\mu(T)$ of vdW Fluid



(Image: He *et al.*, ***Nuclear Physics B* 915**: 243-261 (2017), Figure 3)

Generalization

Use thermodynamics of chemical potentials

to study general properties of phase transitions

1. Phase transitions on (T, p)-space
2. Effect of external pressure on vapor pressure
3. Location of phase boundaries

Use of Gibbs-Duhem Equation

Recall

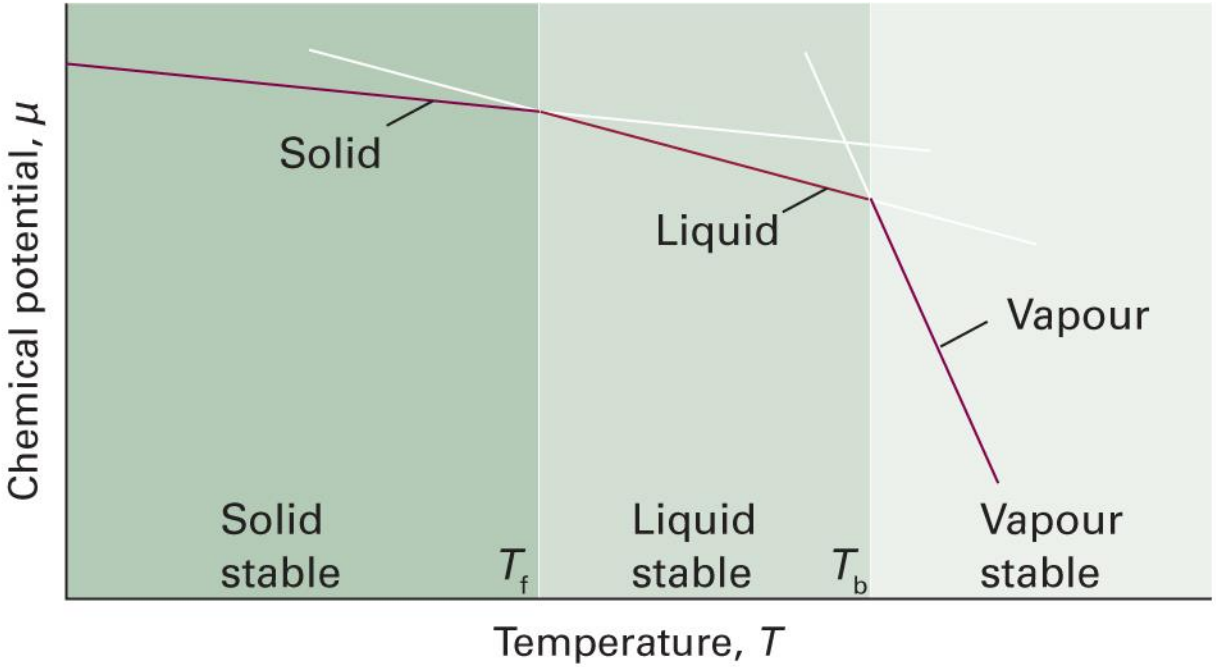
$$\mathrm{d}\mu = -S_\mathrm{m}\mathrm{d}T + V_\mathrm{m}\mathrm{d}p$$

which gives

$$\left(\frac{\partial \mu}{\partial T}\right)_{p,n} = -S_\mathrm{m}, \qquad \left(\frac{\partial \mu}{\partial p}\right)_{T,n} = V_\mathrm{m}$$

Chemical Potential and T

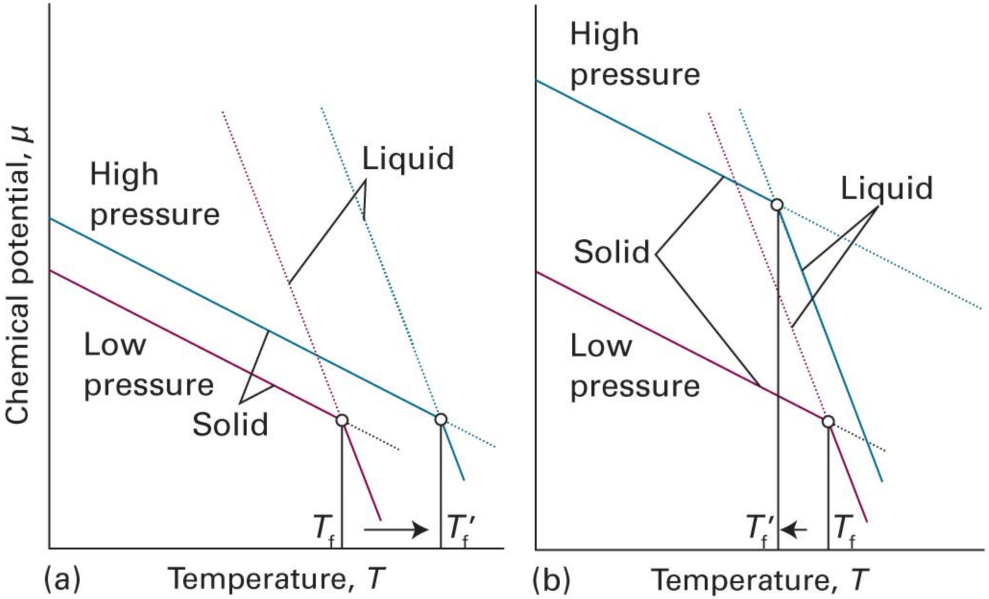
$$\left(\frac{\partial \mu}{\partial T}\right)_{p,n} = -S_{\mathrm{m}}$$



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figures 4B.1)

Chemical Potential and p

$$\left(\frac{\partial \mu}{\partial p}\right)_{T,n} = V_{\mathrm{m}}$$



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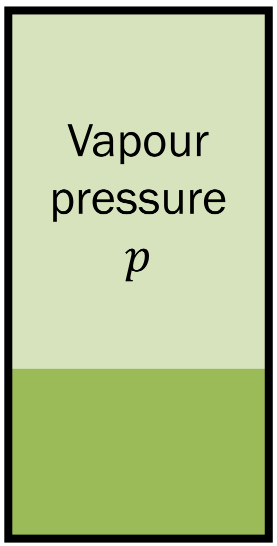
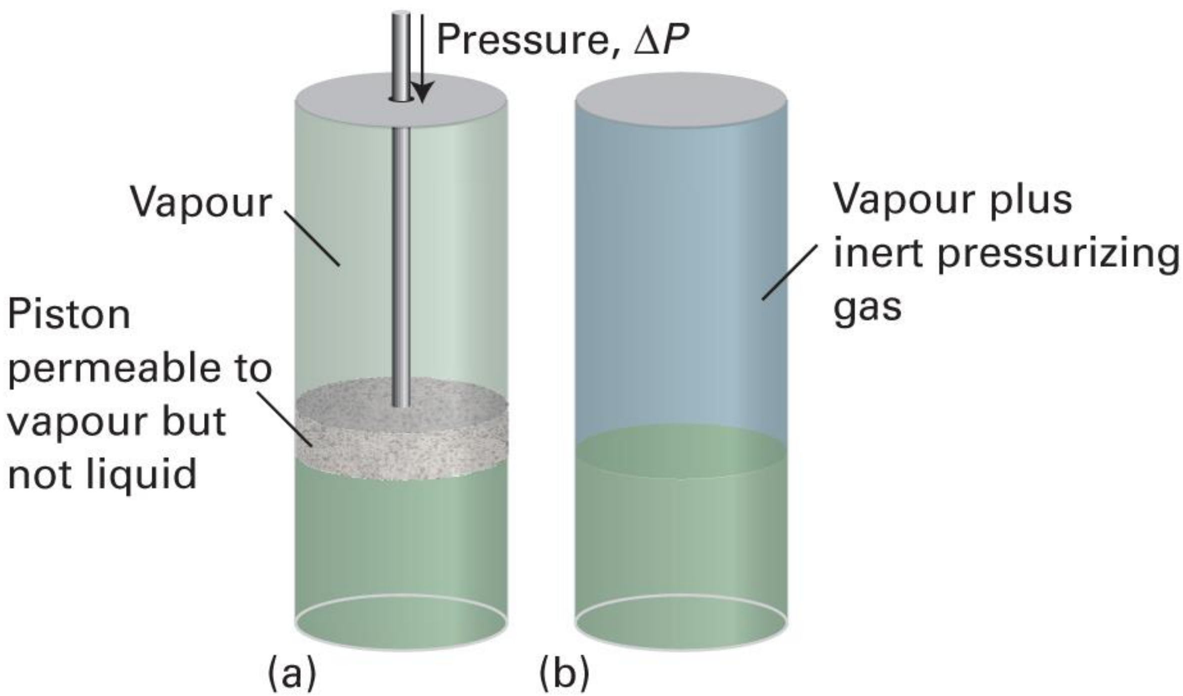
Pressure on Liquid-Gas Eq.

Liquid + Gas





(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figures 4B.3)

Liquid-Gas Equilibrium

At equilibrium,

$$\mu(\mathrm{l}) = \mu(\mathrm{g})$$

$$\mathrm{d}\mu(\mathrm{l}) = \mathrm{d}\mu(\mathrm{g})$$

Since $\mathrm{d}\mu = V_{\mathrm{m}}\mathrm{d}p$,

- Liquid: $\mathrm{d}\mu(\mathrm{l}) = V_{\mathrm{m}}(\mathrm{l})\mathrm{d}P$ (P: external pressure)
- Gas: $\mathrm{d}\mu(\mathrm{g}) = V_{\mathrm{m}}(\mathrm{g})\mathrm{d}p = (RT/p)\mathrm{d}p$ (p: vapor pressure)

$$\frac{RT}{p}\mathrm{d}p = V_{\mathrm{m}}(\mathrm{l})\mathrm{d}P$$

Integration

$$\frac{RT}{p}\,\mathrm{d}p = V_{\mathrm{m}}(\mathrm{l})\mathrm{d}P$$

$$\int_{\text{state } i}^{\text{state } f} \frac{RT}{p}\,\mathrm{d}p = \int_{\text{state } i}^{\text{state } f} V_{\mathrm{m}}(\mathrm{l})\,\mathrm{d}P$$

- Initial state: no external pressure ($P = p = p^*$)
- Final state: $P = p^* + \Delta P$, $p = p$

$$\int_{p^*}^{p} \frac{RT}{p'}\,\mathrm{d}p' = \int_{p^*}^{p^*+\Delta P} V_{\mathrm{m}}(\mathrm{l})\,\mathrm{d}P$$

Effect of ΔP on Vapor Pressure

$$\int_{p^*}^{p} \frac{RT}{p'}\,\mathrm{d}p' = \int_{p^*}^{p^*+\Delta P} V_\mathrm{m}(\mathrm{l})\,\mathrm{d}P$$

Assuming $V_\mathrm{m}(\mathrm{l})$ is constant,

$$RT\ln\left(\frac{p}{p^*}\right) = V_\mathrm{m}(\mathrm{l})\Delta P$$

Therefore,

$$p = p^* e^{V_\mathrm{m}(\mathrm{l})\Delta P/RT}$$